PRECAST PLUGIN

Help Guide & Standard Operating Procedure

Version 1.0 | May 2026

Water Tank	Drain	Wall	BOQ	Bar Bending

Table of Contents

1. Plugin Overview

The **Precast Plugin** is a CAD/BIM productivity tool that automates the generation of parametric 3-D models, construction drawings, Bill of Quantities (BOQ), and bar-bending schedules for precast concrete elements. It eliminates repetitive manual modelling by allowing engineers to select an element type, configure key parameters, and generate fully annotated output in a single click.

Plugin Name	Precast Plugin
Current Version	1.0
Supported Elements	Water Tank, Drain, Wall
Output Types	3-D Model, Drawing, BOQ, Bar Bending Schedule
Target Users	Structural / Civil Engineers, Draughtsmen

2. Interface Guide

2.1 Tab Bar

The top of the dialog contains four tabs that control what action is performed:

Tab	Purpose
Precast	Select element type and configuration. Always start here.
Create Drawing	Generate annotated 2-D construction drawing from the active model.
Update BOQ	Recalculate and export the Bill of Quantities.
Update Bar Bending Schedule	Refresh the bar-bending schedule linked to the model.

2.2 Element Type Panel

After selecting the **Precast** tab, choose the element type by clicking one of the three buttons: **Water Tank**, **Drain**, or **Wall**. The active selection is highlighted in blue.

2.3 Configuration Sub-Panel

Once an element type is chosen, a configuration sub-panel appears below. For **Water Tank**, the options are:

- Monolithic Tank – cast-in-place single-unit construction
- PT Tank – post-tensioned precast tank
- Custom PT Tank – post-tensioned tank with user-defined geometry

After selecting the tank type, choose the **Monolithic Capacity**:

- 6 KL – Standard small residential / commercial
- 10.5 KL – Medium capacity
- 12.5 KL – Large capacity

2.4 Generate Model Button

Click **Generate Model** to create the 3-D parametric model in the active workspace. The button may display a **Locked** badge if the current configuration requires a licence upgrade or if mandatory parameters are incomplete.

■ *The status bar at the bottom of the dialog always reflects the current selection (e.g. 'Type: Monolithic Tank — Standard configuration selected').*

3. Element Types

Element	Description	Sub-types
Water Tank	Underground or overhead precast water storage structures.	Monolithic, PT, Custom PT
Drain	Precast drain channels, box culverts, and kerb units.	(configured in next release)
Wall	Precast retaining walls and boundary wall panels.	(configured in next release)

4. Standard Operating Procedure (SOP) – General

Follow these steps for **every** new precast element generation task. Deviations must be approved by the project lead.

Step	Action / Description	Expected Result
S1	Open the host application (CAD / BIM software) and load or create the project file.	Project file is active in the workspace.
S2	Launch the Precast Plugin from the plugin menu or ribbon.	Plugin dialog opens; status bar shows 'Ready'.
S3	Verify the **Precast** tab is active (default on launch).	Precast tab highlighted; Element Type panel visible.
S4	Select the required **Element Type** (Water Tank / Drain / Wall).	Selection highlighted; Configuration sub-panel appears.
S5	Choose the appropriate **sub-type and capacity / size**.	Status bar updates to confirm selection.
S6	Click **Generate Model**.	3-D model inserted into workspace; no error dialogs.
S7	Switch to the **Create Drawing** tab and generate the drawing.	2-D drawing sheet created with annotations.
S8	Switch to **Update BOQ** tab and export the BOQ.	BOQ exported to project folder as .xlsx or .csv.
S9	Switch to **Update Bar Bending Schedule** and refresh.	Bar bending schedule updated; check rebar counts.
S10	Save the project file. Archive the generated documents per project SOP.	File saved; documents stored in correct location.

■ *Always run Steps 7 – 9 after each model generation to keep drawings and schedules in sync with the 3-D model. Do NOT manually edit the generated BOQ or bar bending schedule.*

5. Detailed Workflow – Water Tank (Monolithic)

This section provides step-by-step guidance for the most common task: generating a **Monolithic Water Tank** model.

5.1 Pre-requisites

- Precast Plugin installed and activated (licence status: Active).
- Host application project file open with correct units (mm) and coordinate system set.
- Structural design parameters confirmed by the engineer (capacity, soil type, loading).

5.2 Step-by-Step

Step	Action / Description	Expected Result
1	Click the **Precast** tab.	Tab active; Element Type buttons visible.
2	Click **Water Tank** in the Element Type panel.	Water Tank highlighted (blue border). Sub-panel shows Monolithic Tank / PT Tank / Custom PT Tank.
3	Click **Monolithic Tank**.	Monolithic Capacity options appear: 6 KL, 10.5 KL, 12.5 KL.
4	Select the required capacity, e.g. **6 KL**.	Status bar: 'Type: Monolithic Tank — Standard configuration selected'.
5	Click **Generate Model**.	Plugin processes; 3-D solid model of the 6 KL tank inserted at the specified insertion point.
6	Inspect the model — verify overall dimensions and wall thickness in the properties panel.	Dimensions match design specs from the calculation sheet.
7	Go to **Create Drawing** tab → click **Create Drawing**.	New drawing sheet with plan, section, and elevation views generated.
8	Review drawing for annotation completeness (title block, scale, revision mark).	All annotations present; title block auto-populated.
9	Go to **Update BOQ** → click **Update BOQ**.	BOQ worksheet opened / updated; quantities reflect model.
10	Go to **Update Bar Bending Schedule** → click **Update**.	Schedule refreshed; rebar count and lengths verified against design.
11	Save project. Issue drawings and schedules for review.	Files saved to project directory; sent to reviewer.

5.3 Acceptance Criteria

Check	Criterion	By
Model geometry	Matches capacity and design thickness	Modeller
Drawing completeness	All views, annotations, and title block present	Draughtsman
BOQ accuracy	Quantities within ±2% of manual estimate	Engineer
Bar Bending Schedule	Rebar count and lengths match structural design	Engineer

6. Troubleshooting

Issue	Likely Cause	Resolution
'Locked' badge on Generate Model	Licence not activated or expired	Contact your administrator to activate / renew the licence key.
Model inserts at wrong position	Insertion point not set	Pre-select the insertion point in the host app before clicking Generate Model.
BOQ shows zero quantities	Model not saved before updating BOQ	Save the project file, then re-run Update BOQ.
Drawing not created	No active drawing sheet in project	Create or open a drawing sheet in the host app first.
Bar bending schedule rebar count mismatch	Model modified after last update	Regenerate the model and re-run Update Bar Bending Schedule.
Plugin dialog does not open	Plugin not installed or DLL missing	Reinstall the plugin. Check logs in %AppData%\PrecastPlugin\logs.

7. Glossary

BOQ	Bill of Quantities – itemised list of materials and quantities.
Bar Bending Schedule	Document listing each reinforcement bar: shape, diameter, length, and quantity.
Monolithic Tank	Water tank cast as a single reinforced concrete unit.
PT Tank	Post-tensioned precast tank where tendons are stressed after casting.
Custom PT Tank	Post-tensioned tank with user-specified non-standard dimensions.
KL	Kilolitre – unit of liquid volume (1 KL = 1,000 litres = 1 m^3).
Precast	Concrete elements manufactured off-site and installed on location.
Generate Model	Plugin action that creates the parametric 3-D solid model.
Locked	Indicates the feature requires licence activation before use.

For support, contact the plugin team or raise a ticket in the internal helpdesk portal.